01-14B FUEL SYSTEM [WL-C, WE-C]

01

FUEL SYSTEM OUTLINE [WL-C, WE-C]

dcf011400000t04

Features

Improved exhaust gas purification	Common rail injection system adopted

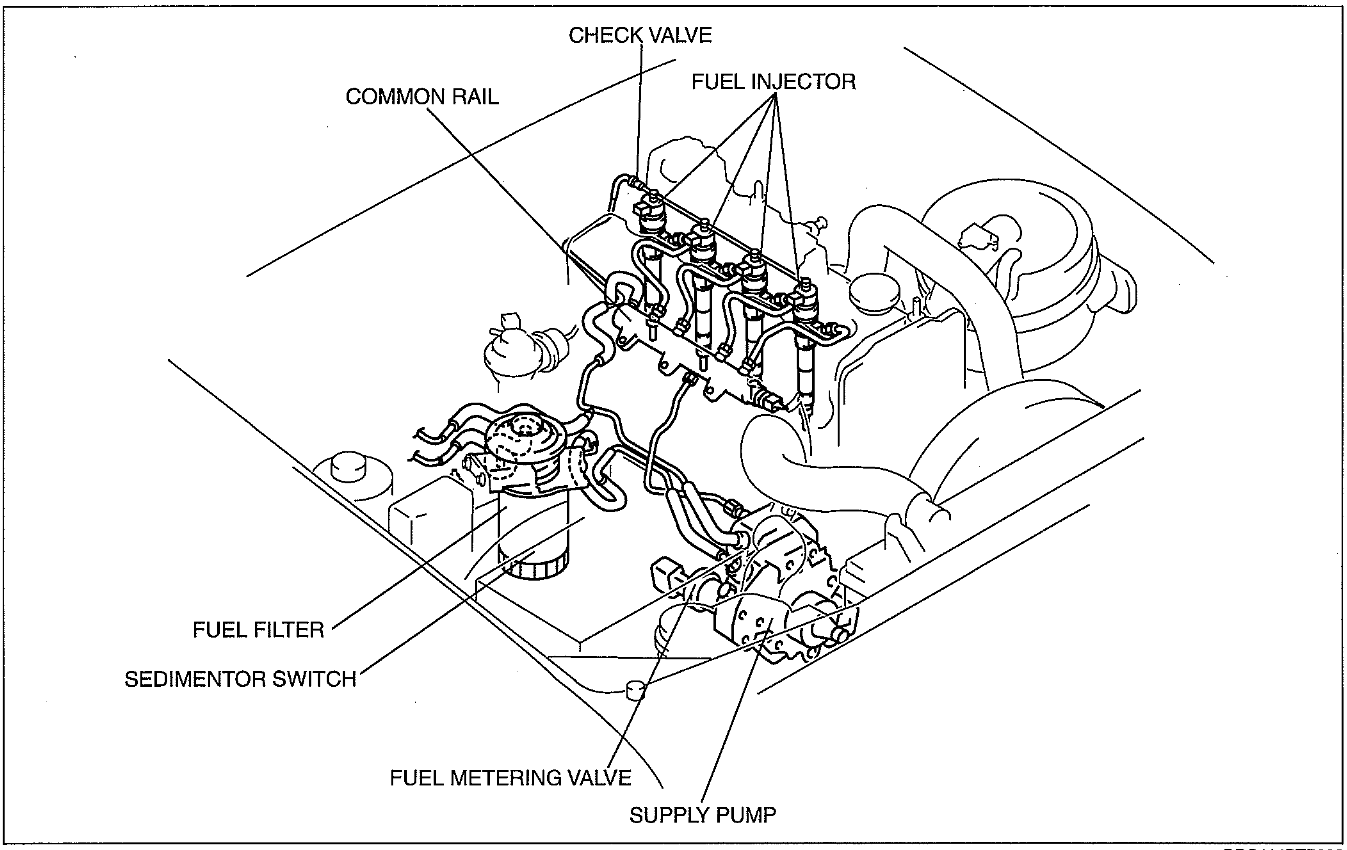
Specification

Item		WL-C, WE-C
Supply pump		Electronic control
Fuel injector		Electromagnetic control
Fuel tank capacity (reference)	(L {US gal, Imp gal})	70 {18, 15}

FUEL SYSTEM STRUCTURAL VIEW [WL-C, WE-C]

dcf011400000t05

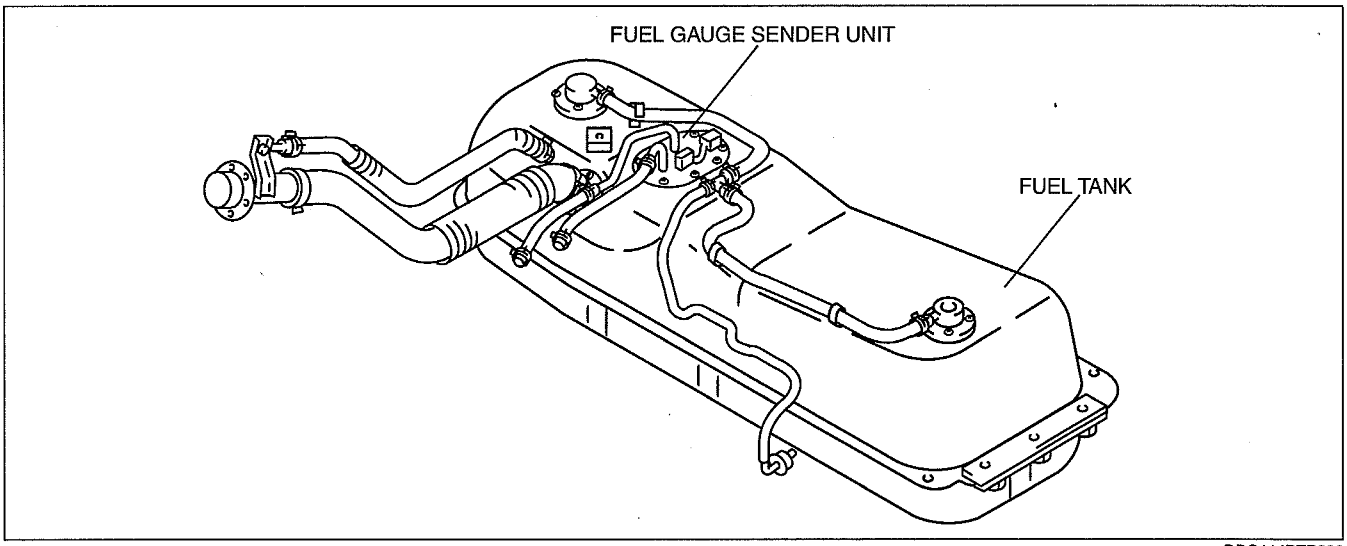
Engine Room Side



DBG114BTB302

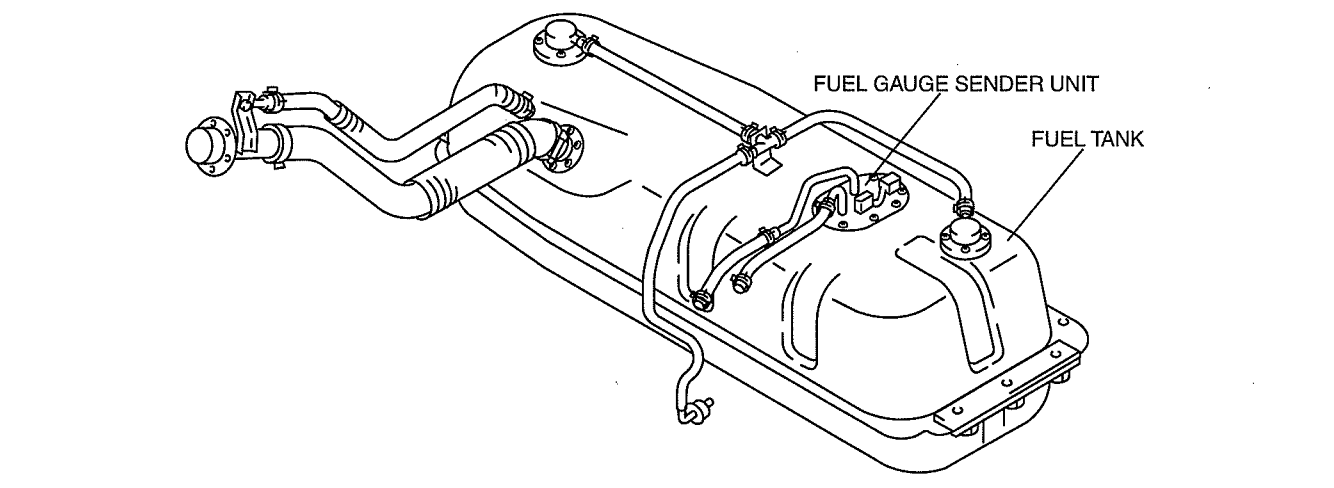
Fuel Tank Side

Double cab 4x2 (except Hi-Rider), Stretch cab 4x2 (with rear access system (except Hi-Rider))



DBG114BTB330

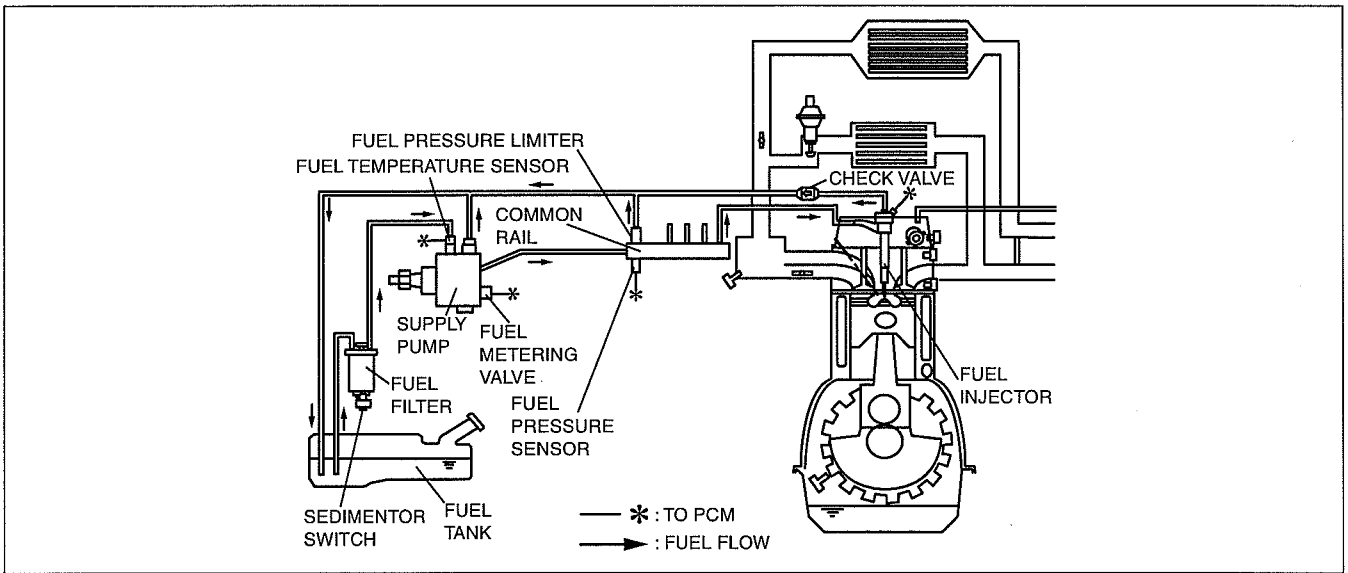
Regular cab 4x4, stretch cab 4x4 (with rear access system), Hi-Rider



DBG114BTB331

FUEL SYSTEM DIAGRAM [WL-C, WE-C]

dcf011400000t06



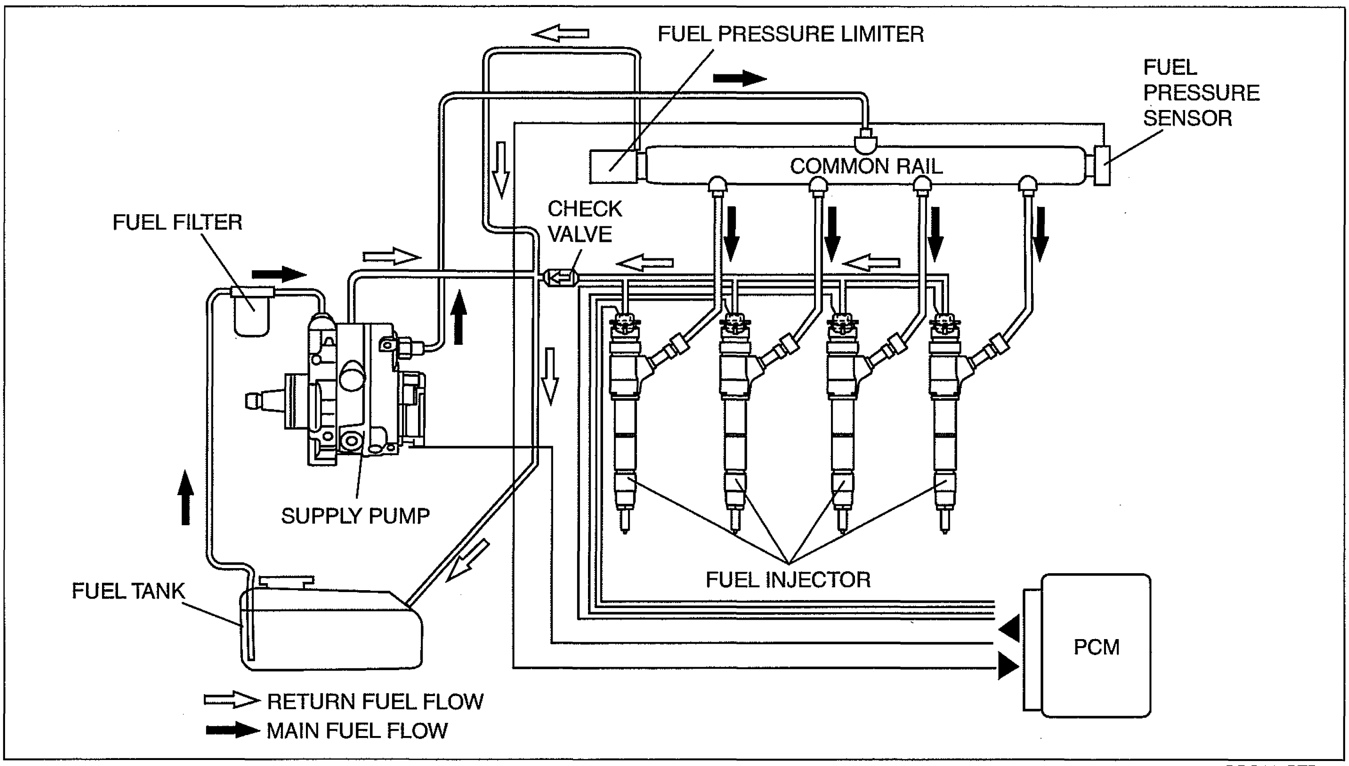
DBG114BTB301

COMMON RAIL INJECTION SYSTEM FUNCTION [WL-C, WE-C]

dcf011400000t07

This system stores fuel under high pressure supplied by the supply pump in the common rail, and injects it through the electromagnetic injectors.
The PCM controls both fuel injection amount and timing by sending a signal to the injector solenoid valve coil.
Since fuel is highly pressurized by the supply pump and stored in the common rail, injection pressure is constantly assured even at low speed since engine speed and load have no influence on the injection process.
The PCM can precisely control the fuel injection amount and timing due to use of an solenoid valve coil in the injector that opens and closes the fuel path.
Fuel is injected multiple times, including multiple auxiliary injections previous to the main injection. These multiple injections help to smooth the fuel combustion process, reducing noise and vibration levels as well as the amount of NOx in the exhaust.

With the use of 160 MPa {1.632 kgf/cm^2, 23,206 psi} ultra-high pressure injectors, engine output is improved and the amount of smoke and particulate matter in the exhaust is greatly reduced.



DBG114BTB308

FUEL TANK CONSTRUCTION [WL-C, WE-C]

dcf011442110t02

Capacity is 70 L {18 US gal, 15 Imp gal}.
Two rollover valves are built-in.

FUEL FILTER FUNCTION [WL-C, WE-C]

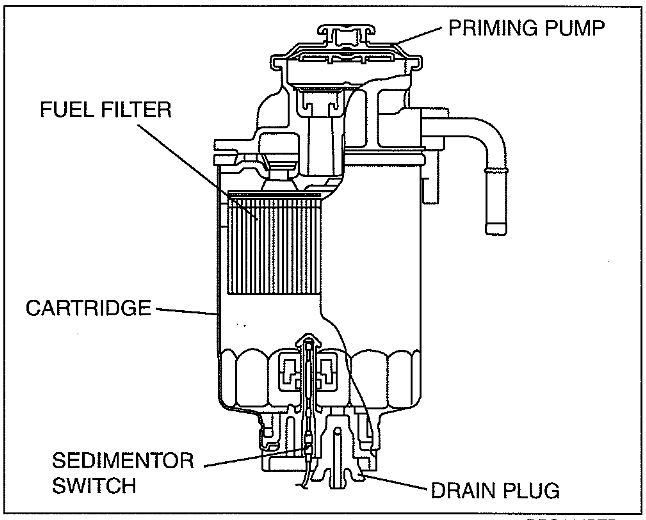
dcf011420490t03

The filter part removes foreign material or dirt in the fuel.
The priming pump is a pump to discharge (drain) the accumulated water efficiently.

FUEL FILTER CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf011420490t04

- The fuel filter consists of the drain plug and the priming pump.
- To drain water, loosen the drain plug on the lower side and pump the priming pump on the upper side of the fuel filter.
- The sedimentor switch illuminates the sedimentor warning light to inform the user of the drain timing. The float rises according to the level of accumulated water. If the water level exceeds the specification, the circuit for the sedimentor switch is completed.



DBG114BTB307

SUPPLY PUMP FUNCTION [WL-C, WE-C]

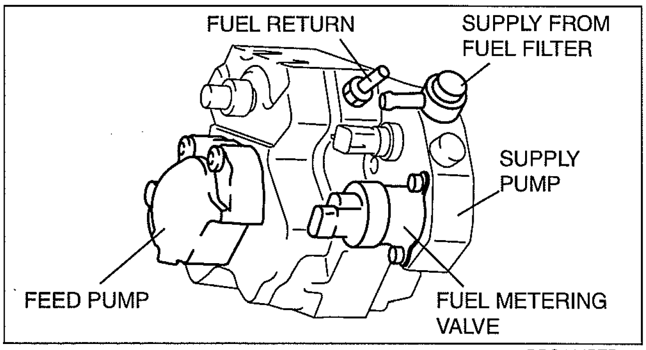
dcf011413350t01

- An electrically controlled supply pump with ultra-high discharge pressure and low-drive torque has been adopted.
- The supply pump intakes the proper fuel amount and pressure feeds the fuel to the common rail.

SUPPLY PUMP CONSTRUCTION/OPERATION [WL-C, WE-C]

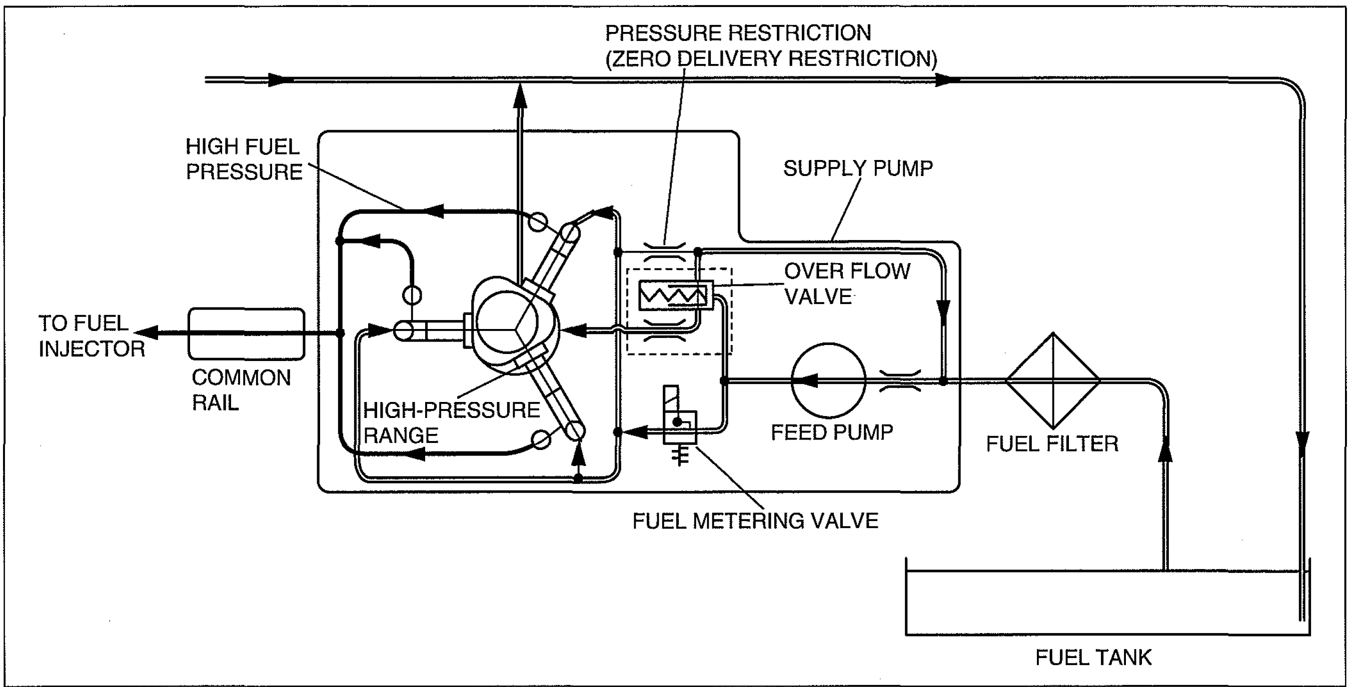
dcf011413350t02

- The supply pump is driven by the timing belt. The feed pump is driven by the supply pump drive shaft.
- The supply pump has three pumping elements with 120° offset (displacement pumps).
- The fuel metering valve is located in the supply duct between the feed pump and the supply pump, and meters the quantity of fuel which is to be delivered into the high-pressure chamber in accordance with the current operating status of the engine.
- Any surplus fuel is returned to the fuel tank via the fuel return.



DBG114BTB502

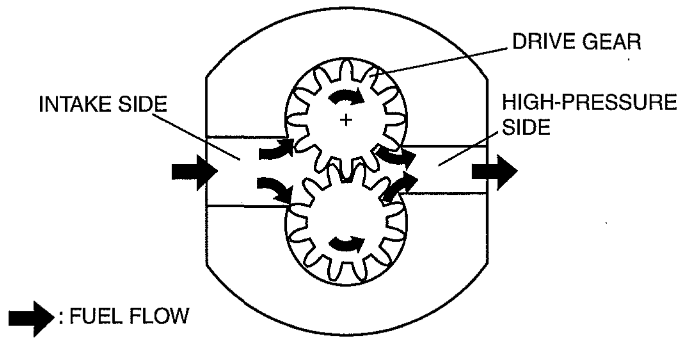
Flow of fuel through fuel pump



DBG114BTB509

Feed Pump Operation

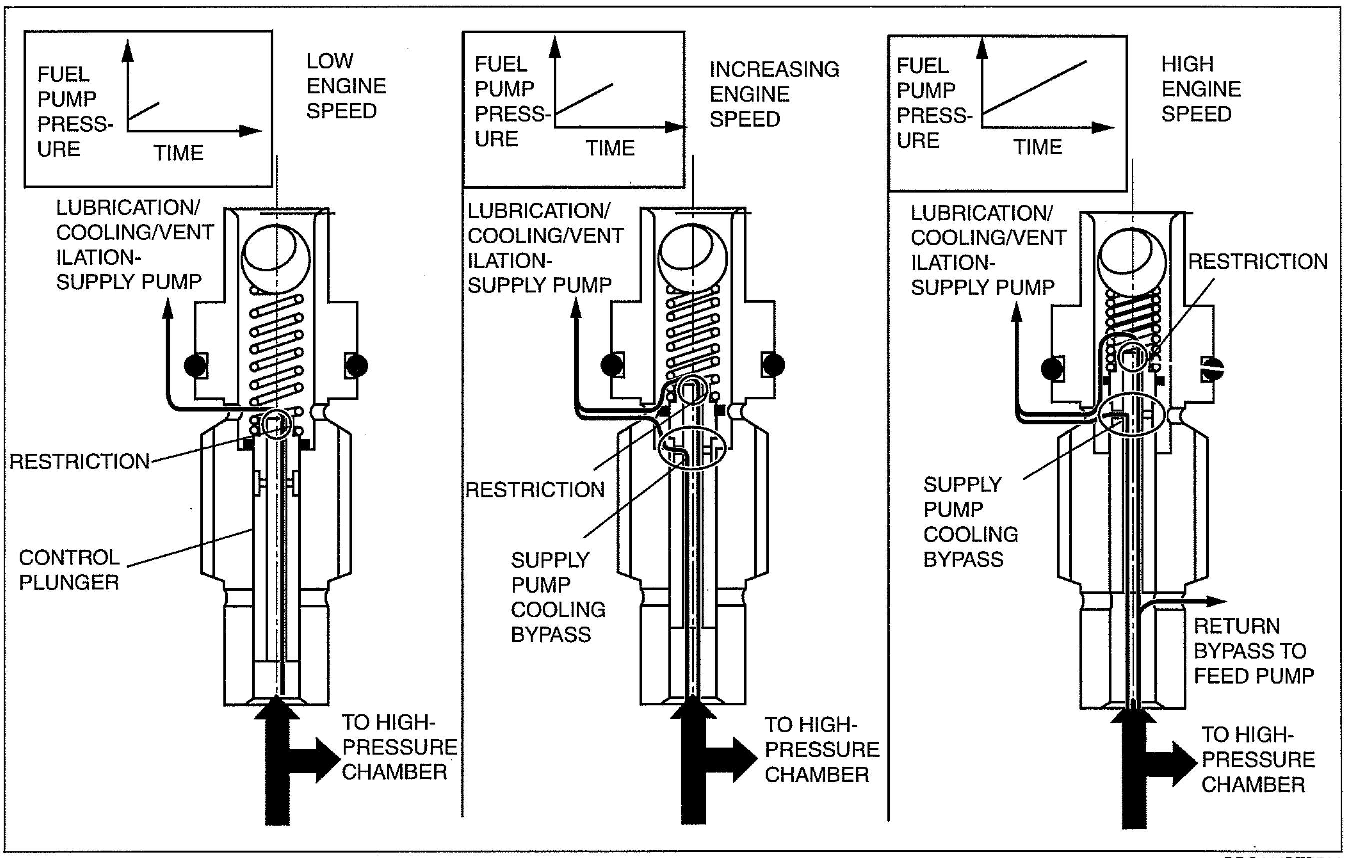
- The feed pump is designed as a gear pump and delivers the required fuel to the supply pump.
  Essential components are two counter-rotating, meshed gear wheels that transport the fuel in the tooth gaps from the intake side to the pressure side.
  The contact line of the gears forms a seal between the intake side and the pressure side and prevents the fuel from flowing back.
  The delivery quantity is approximately proportional to engine speed. For this reason, fuel-quantity control is required.
  For fuel-quantity control purposes, there is an overflow valve incorporated in the supply pump.



DBG114BTB510

Overflow Valve Operation

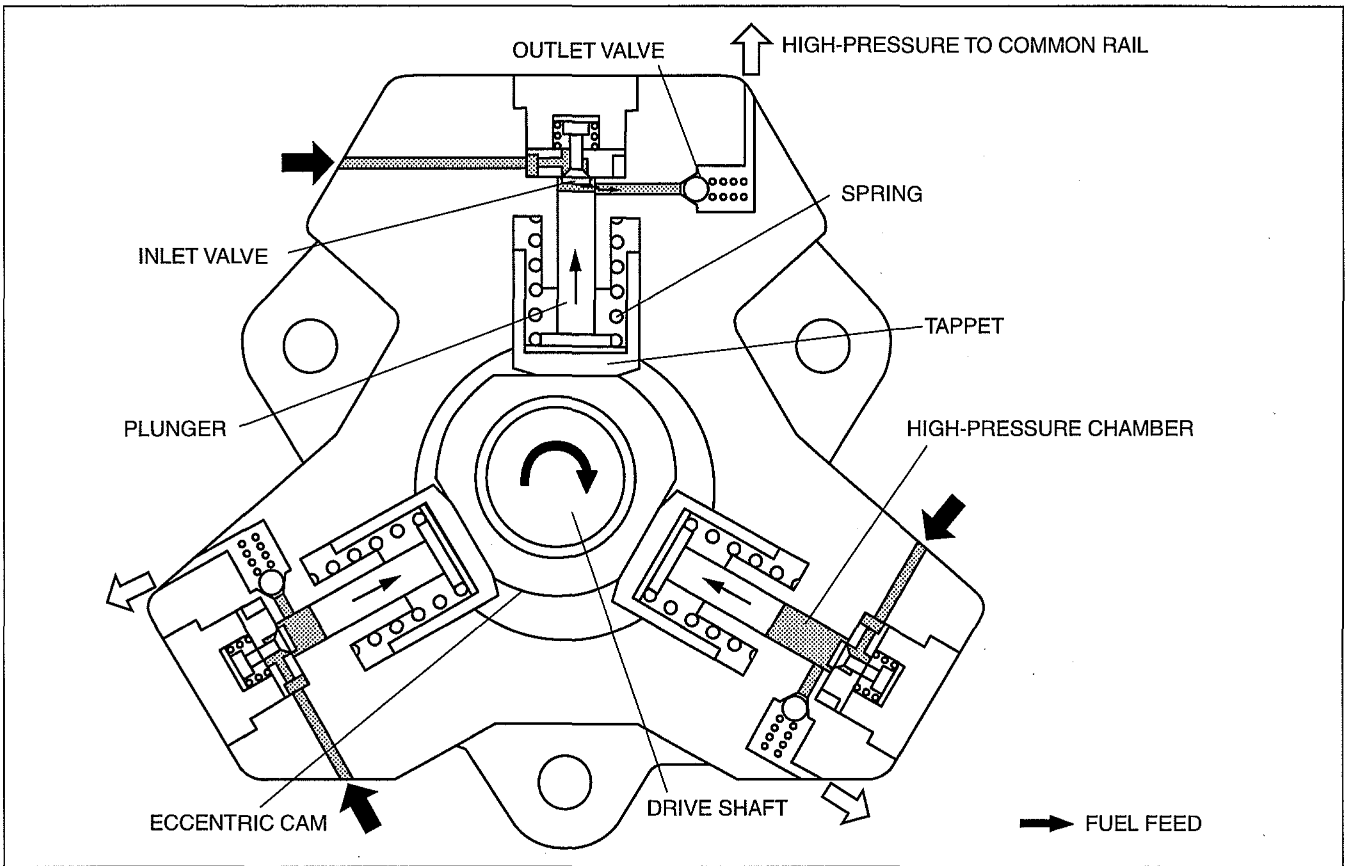
- High-pressure generation (up to 160 MPa {1,632 kgf/cm^2, 23,206 psi}) means high thermal load on the individual components of the supply pump. The mechanical components of the supply pump must also be lubricated sufficiently to ensure durability.
  The overflow valve is designed to ensure optimum lubrication or cooling for all operating conditions.
  At lower engine speeds (low feed pump pressure) the control plunger is moved only slightly out of its seat. The lubrication/cooling requirement is correspondingly low.
  A small amount of fuel is released to lubricate/cool the pump via the restriction at the end of the control plunger The supply pump features automatic venting. Any air in the supply pump is vented through the restriction.
  With increasing engine speed (increasing feed pump pressure), the control plunger is moved further against the compression spring.
  Increasing engine speeds require increased cooling of the supply pump. Above a certain pressure, the supply pump cooling bypass is opened and the flow rate through the supply pump is increased.
  At high engine speeds (high feed pump pressure), the control plunger is moved further against the compression spring. The supply pump cooling bypass is now fully open (maximum cooling).
  Excess fuel is transferred via the return bypass to the intake side of the feed pump.
  In this way, the internal pump pressure is limited to a maximum of 600 kPa {6.12 kgf/cm^2, 87.0 psi}.



DBG114BTB511

High Pressure Generation

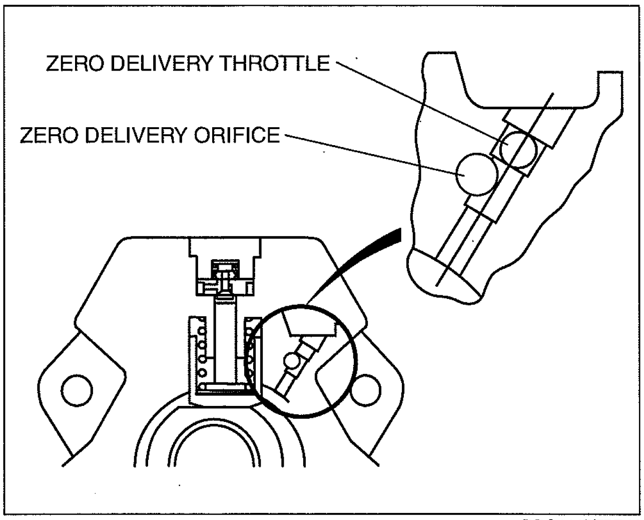
- The supply pump is driven via the drive shaft.
  An eccentric element is fixed to the drive shaft and moves the three plungers up and down according to the cam lobes of the eccentric element.
  Fuel pressure from the feed pump is applied to the inlet valve. If the feed pump pressure exceeds the internal pressure of the high-pressure chamber (pump plunger in TDC position), the inlet valve opens.
  Fuel is now pressed into the high-pressure chamber, which moves the pump plunger downwards (intake stroke).
  If BDC of the pump plunger is exceeded, the inlet valve closes due to the increasing pressure in the high-pressure chamber. The fuel in the high-pressure chamber can no longer escape.
  As soon as the pressure in the high-pressure chamber exceeds the pressure in the common rail, the outlet valve opens and the fuel is pressed into the common rail via the high-pressure connection (delivery stroke).
  The pump plunger delivers fuel until TDC is reached. After this, the pressure drops so that the outlet valve closes.
  As the pressure on the remaining fuel is reduced, the pump plunger moves downward.
  If the pressure in the high-pressure chamber falls below the transfer pressure, the inlet valve reopens and the process starts again.



DBG114BTBR12

Zero Delivery Restriction Operation

- The zero-delivery throttle is installed to the supply pump. The fuel metering unit sends a quantity of fuel to the plunger even during zero-delivery. To prevent this, the fuel is returned to the feed pump inlet via the orifice of the zero-delivery throttle, ensuring zero fuel delivery to the plunger. Except during zero-delivery, essentially no fuel is sent to the orifice due to the throttle effect, instead it is sent to the plunger.

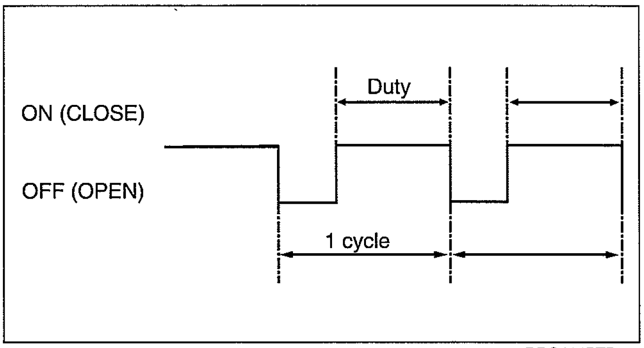


DBG114BTBR13

FUEL METERING VALVE FUNCTION [WL-C, WE-C]

dcf011413350t03

- By controlling the supply pump intake amount, the fuel metering valve controls the pressure pump feed amount and the pressure inside the fuel injection pipe.
- The intake amount is determined by the total opening size of the valve as controlled by the duty value of the solenoid actuation current.
- The duty solenoid can adjust the open and close time ratio to control the intake amount by changing the OFF time during a single cycle. For a normally open type fuel metering valve, making the duty ratio larger increases the intake amount and, conversely, making the duty value smaller decreases the intake amount.



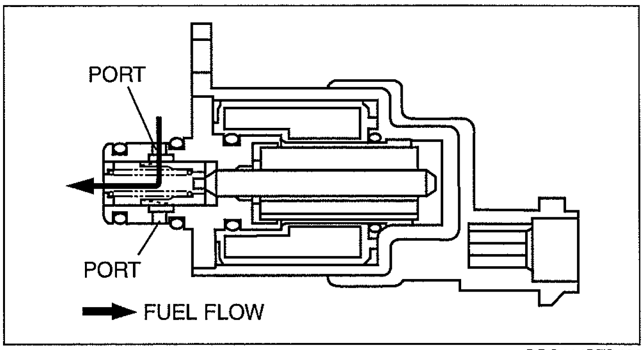
DBG114BTB514

FUEL METERING VALVE CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf011413350t04

De-energized (open)

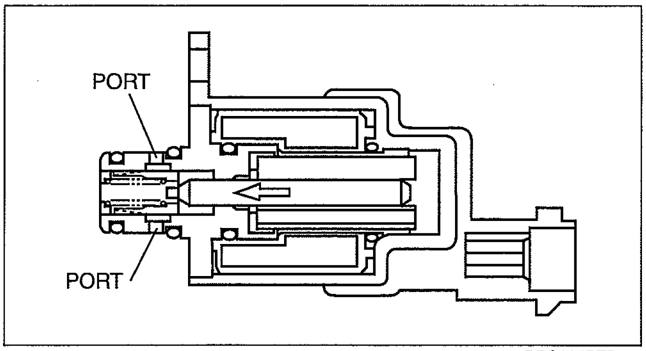
- The solenoid inner port is opened and fuel is sent into the high-pressure chamber.



DBG114BTB506

Energized (closed)

- The solenoid inner port is closed. Fuel siphoned by the feed pump is returned to the fuel tank through the overflow valve.



DBG114BTB507

COMMON RAIL FUNCTION [WL-C, WE-C]

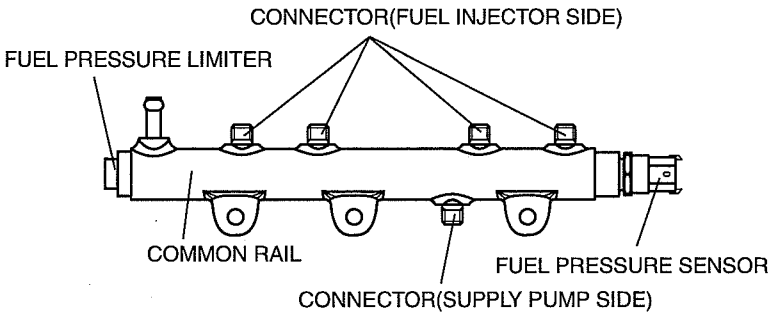
dcf011413151t01

- Due to constant high-pressure fuel (25—160 MPa {255—1.631 kgf/cm^2, 3,626—23,206 psi}) in the rail, electrically controlled injection is made possible.

COMMON RAIL CONSTRUCTION [WL-C, WE-C]

dcf011413151t02

- The PCM constantly maintains optimum pressure by monitoring the fuel pressure measurement signal from the fuel pressure sensor.
- As a consideration to safety, a fuel pressure limiter has been installed to mechanically drain fuel if the pressure goes above 195 MPa {1.988 kgf/cm^2, 28,282 psi}.



DBG114BTB505

FUEL INJECTOR FUNCTION [WL-C, WE-C]

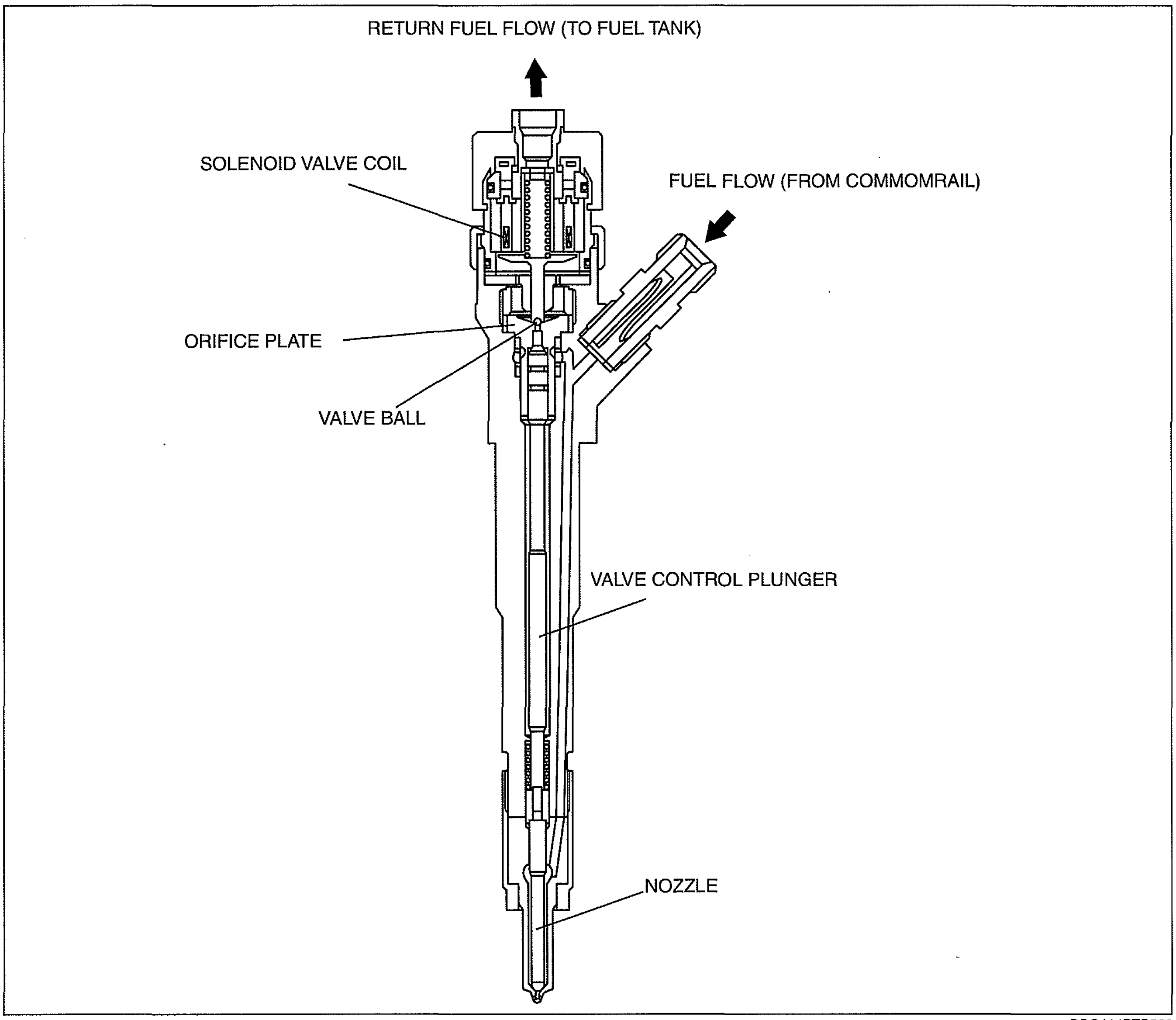
dcf011413250t01

- A small sized, energy saving, electromagnetically controlled injector has been adopted.
- A multiple fuel injection system has been adopted. The fuel injection amount and timing, as well as the division of the numerous injections are according to a PCM ON/OFF signal that controls the opening and closing of the electromagnetic valves.
- The construction and material of the electromagnetic valve has been modified to save energy and improve response.

FUEL INJECTOR CONSTRUCTION [WL-C, WE-C]

dcf011413250t02

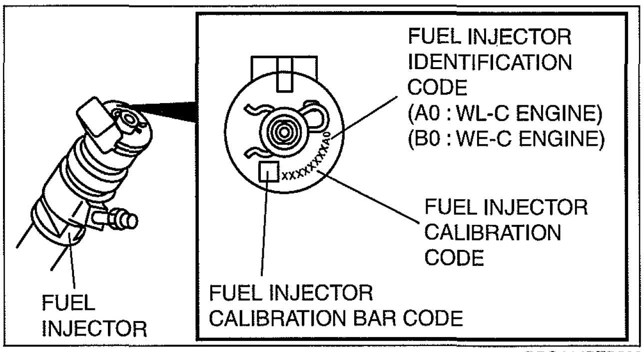
- The fuel injector is composed of: a valve control plunger that controls the nozzle, a control chamber with an inflow/outflow orifice, and an solenoid valve coil that controls the inflow/outflow of fuel to the control chamber via an on/off function.
- The valve control plunger is connected to the nozzle so that the up and down movement of the plunger results in the opening and closing of the nozzle.
- The control chamber contains a single orifice plate to provide an inflow and outflow path.



DBG114BTB500

Fuel Injector Calibration Code

- Because the fuel injector corrects the deviation of the fuel injection amount caused by the difference in mechanical characteristics, if the fuel injector or the PCM is replaced, the injector calibration code (rank number) programming in the PCM is required.
- This is done by inputting the eight digit fuel injector calibration code (or fuel injector calibration bar code) into the PCM by means of M-MDS and taking into account the corresponding cylinder.
- It is not necessary to configure the 9th and 10th digit (A0: WL-C engine, B0: WE-C engine) of the fuel injector calibration code as it is the fuel injector grade identification code.



DBG114BTB503

FUEL INJECTOR OPERATION [WL-C, WE-C]

dcf011413250t03

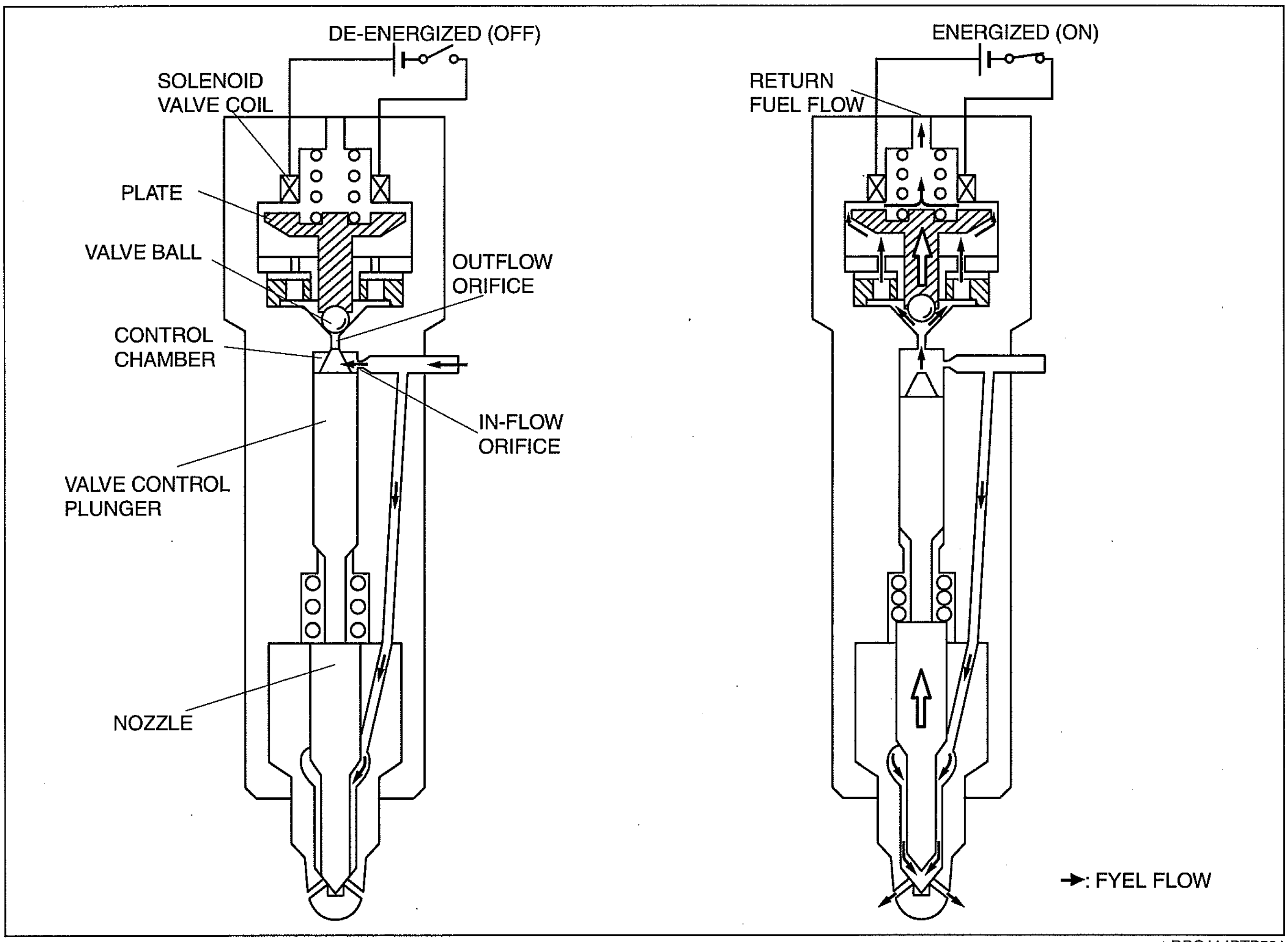
- The solenoid valve coil controls inflow/outflow of fuel to the control chamber by opening and closing according to an ON/OFF signal received from the PCM.

De-energized (off)

— When the solenoid valve coil is not energized, the plate is pushed down, closing the control chamber outflow orifice to valve ball, and causing the pressurized fuel from the fuel injection pipe to fill the control chamber. Due to this, the nozzle connected to the valve control plunger is closed and injection does not occur.

Energized (on)

— When solenoid valve coil energization begins the valve ball and plate are pulled upwards, opening the control chamber outflow orifice and fuel outflow occurs. As fuel outflow occurs, the control chamber pressure decreases, drawing the valve control plunger and the nozzle upward, and injection begins. As energization continues the nozzle is opened even more, providing maximum injection rate. When energization of the solenoid valve coil ends, the plate once again moves down and the nozzle is closed instantaneously. Since the coil can be repeatedly energized any number of times, multiple pilot injection is made possible.

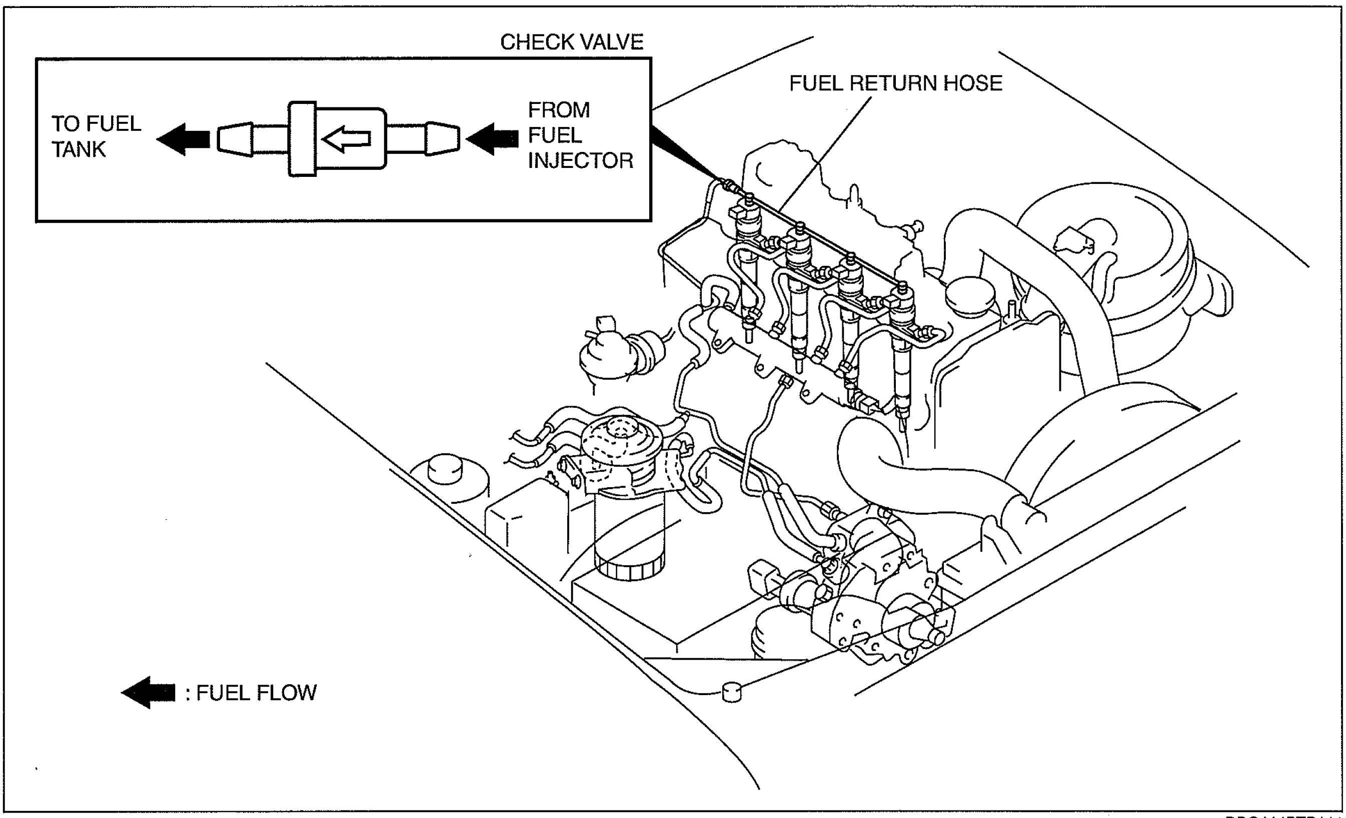


DBG114BTB501

CHECK VALVE FUNCTION [WL-C, WE-C]

dcf011413152t01

- The one-way check valve prevents fuel from flowing back to the fuel injector.



DBG114BTB111